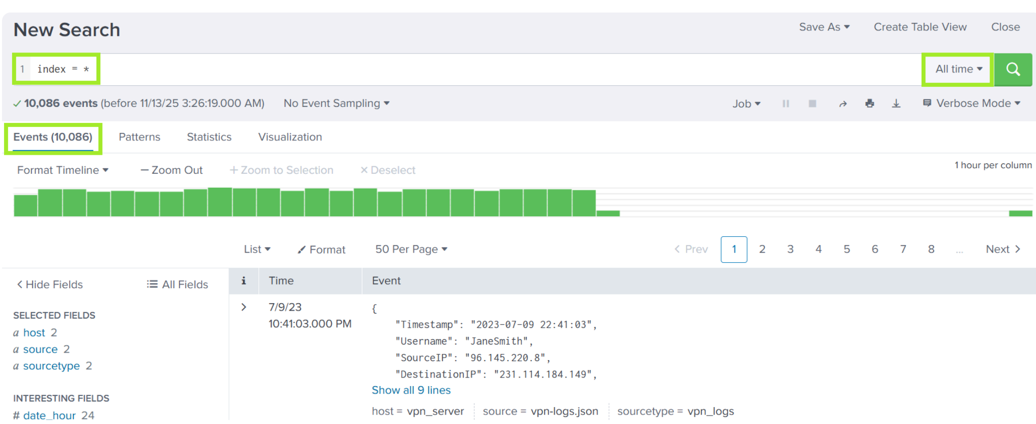
L2-2.3. Splunk: Dashboards and Reports

Task 1 - Introduction

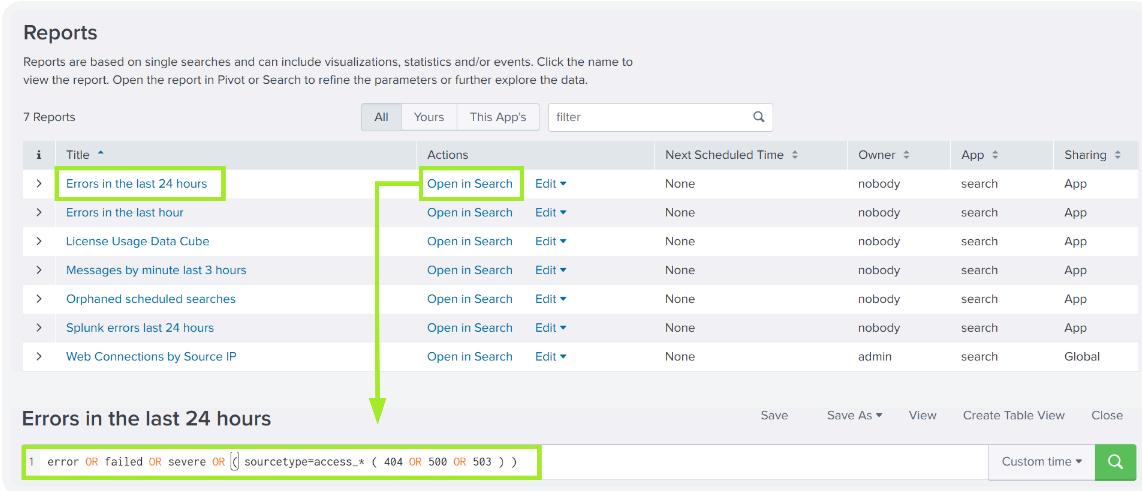
Splunk is one of the most widely used Security Information and Event Management (SIEM) solutions in enterprise environments. It helps aggregate data from various data sources within an enterprise environment to enhance security monitoring. However, large volumes of data can quickly overwhelm analysts. In this room, you'll learn practical ways to organize, visualize, and manage data in Splunk to make analysis faster, clearer, and more effective.

Task 2 - Creating Reports for Recurring Searches

Once Splunk has loaded, head to the **Search & Reporting** app to enter your first query `index = *` and set the time range to `All time` so you can get an overview of the data we will be utilizing in this lab.



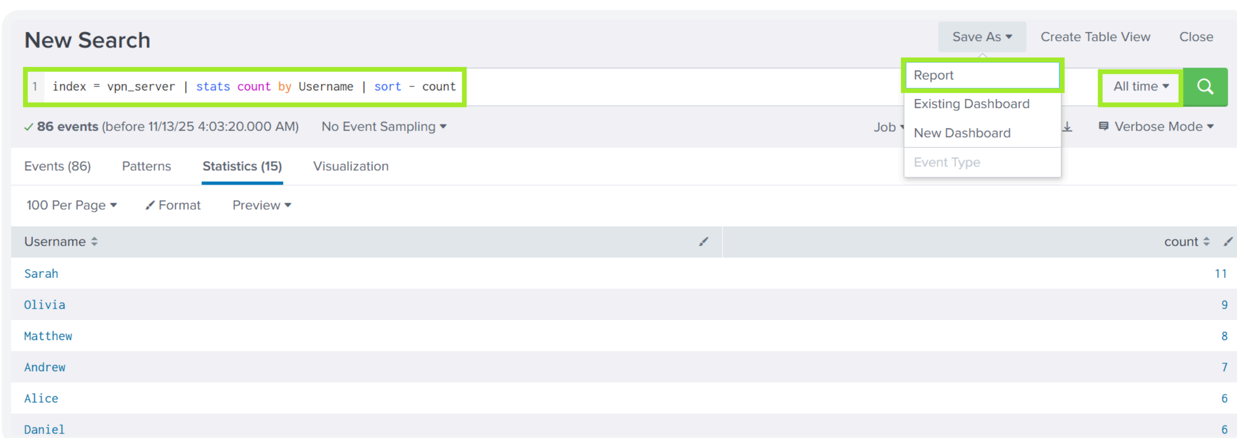
Building Reports: Heading to the **Reports** tab in Splunk click `Open in Search` to view the query and time settings of the report in the search app.



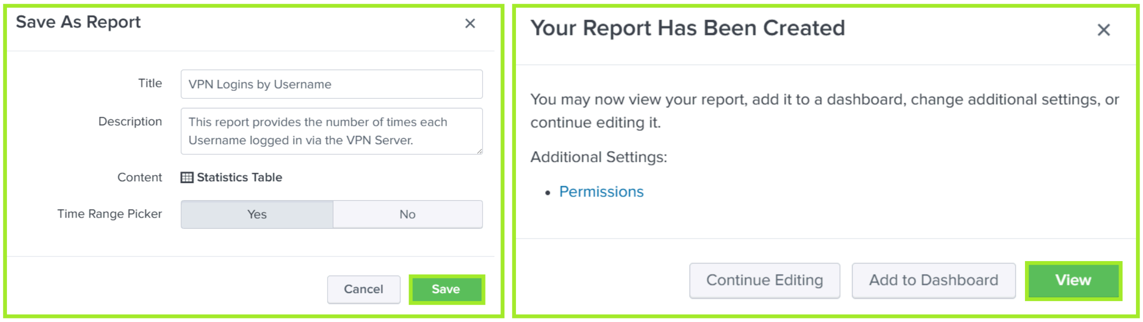
Set the time range to `All time`.

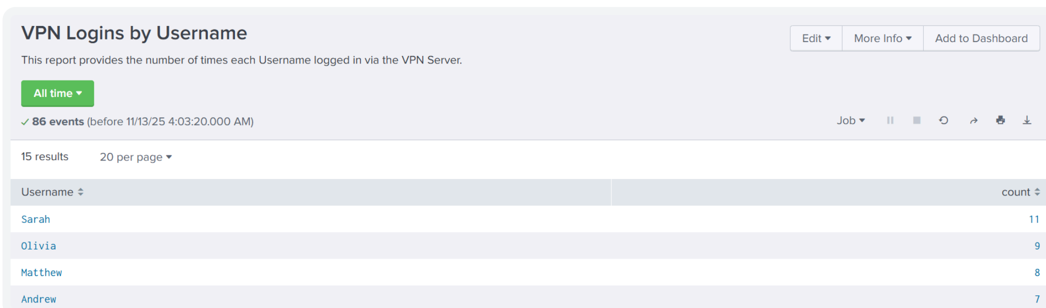
```
index = vpn_server | stats count by Username | sort - count
```

This query will search our `vpn_server` index, give us a count of events based on the `Username` field, and sort them based on the `count` of events. You can see in your Splunk instance and in the screenshot below that we have a total of 86 events, and `Sarah` has the most events recorded. Go ahead and click `Save As` → `Report`.

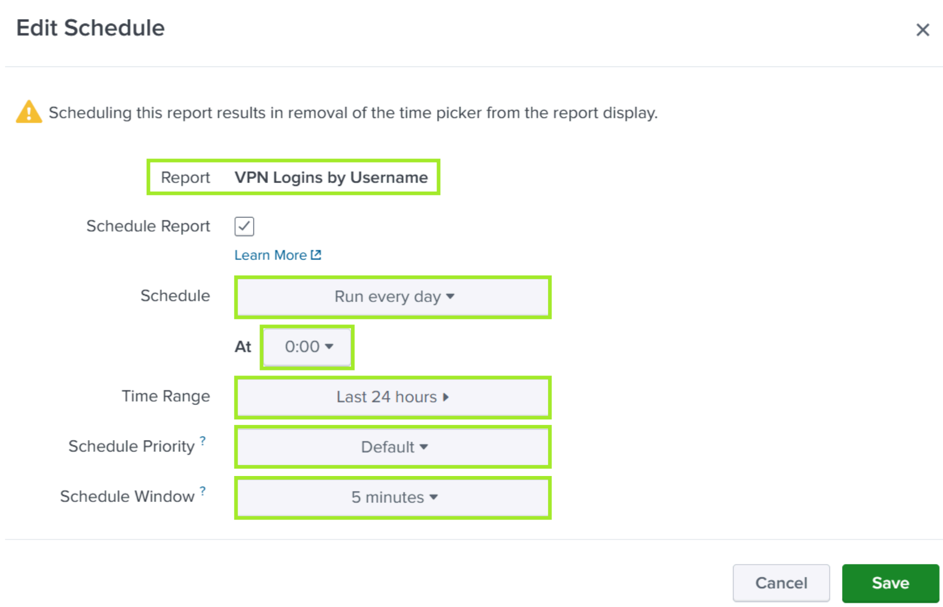


In a SOC environment, you may want to track the users who logged in during a specific time period. Creating a report from the previous query that runs every few hours or daily would be extremely helpful. Let's give our new report a title and description. We used `stats count` in our query, so the `Content` type is automatically set to `Statistics Table`. You can also choose whether to make the `Time Range Picker` active or inactive in your report. `Save` and `View`.





Scheduling a Report: Splunk Enterprise unlocks full scheduling capabilities. As shown in the screenshot below, you can configure reports to run on a recurring schedule, such as daily at midnight, using the previous 24 hours of data. Enterprise also allows you to define a report's priority and scheduling window, enabling Splunk to coordinate multiple scheduled reports efficiently and prevent resource usage issues.



stats count by Username
| sort -count

This is one of the most commonly used SPL commands for creating quick summaries and identifying top users, hosts, IPs, processes, or other fields.

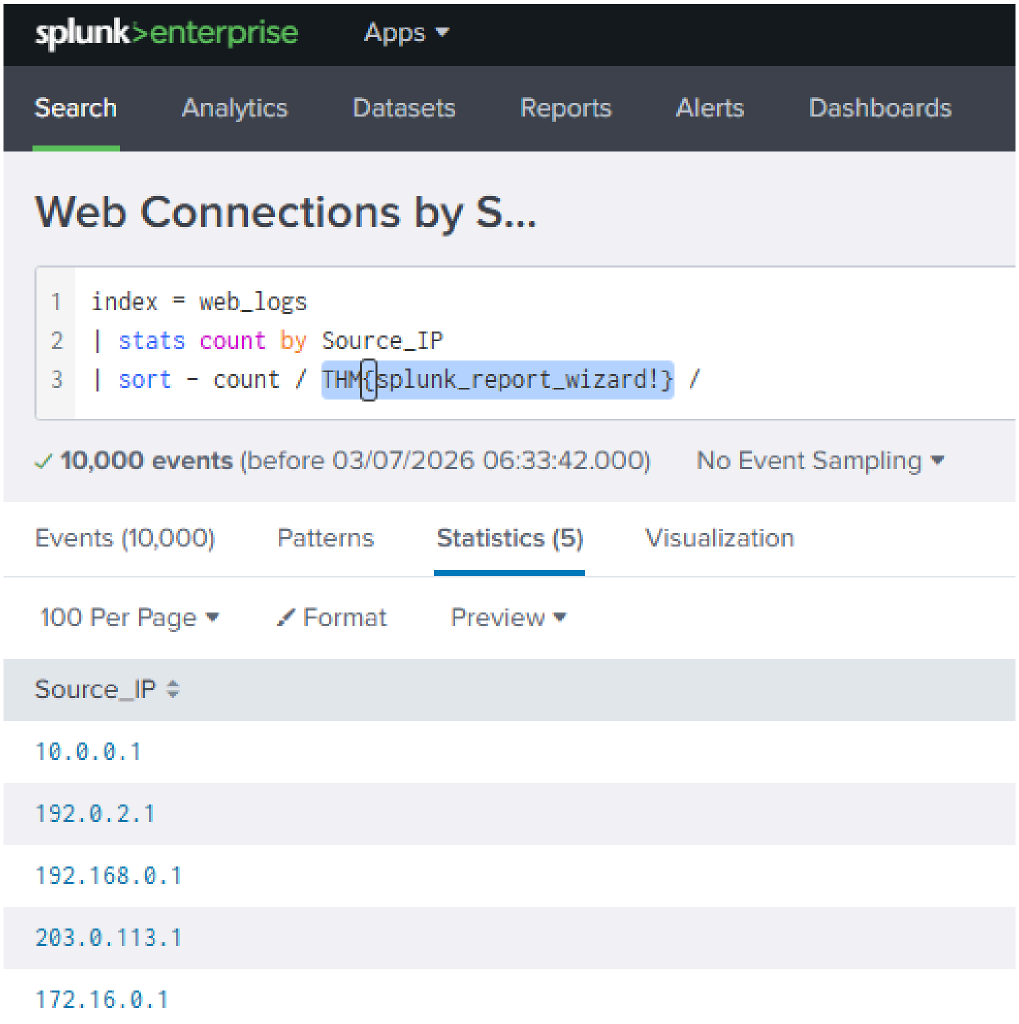
Answer the questions below

> Head to the Splunk Reports tab and open the Web Connections by Source IP report. Which Source_IP field value recorded the highest number of events?

10.0.0.1

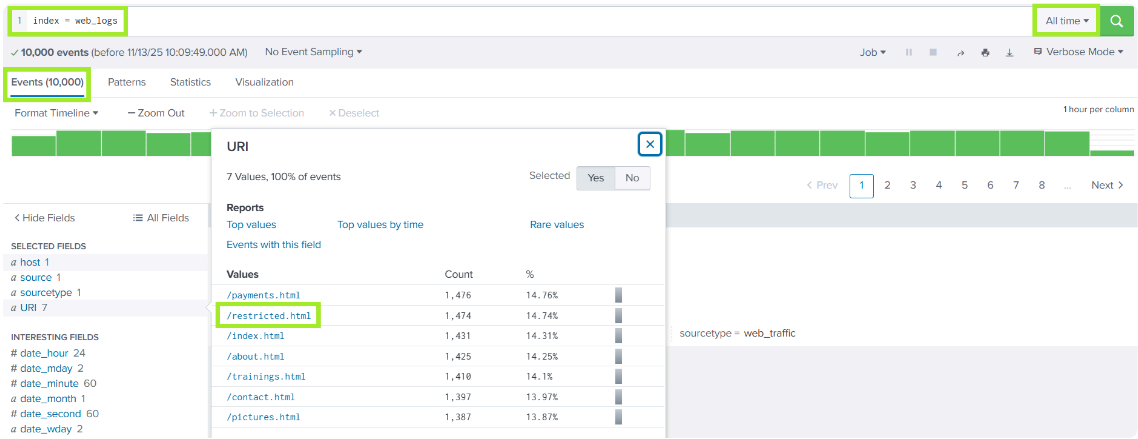
> While viewing the report above, click Edit → Open in Search to investigate the query behind the report. What is the flag value hidden within the search query?

THM{splunk_report_wizard!}



Task 3 - Detecting With Alerts and Rules

Building Alerts: Let's jump into the `web_logs` index and review the available data. We have 10,000 logs available for analysis. Checking out the `URI` field, `/restricted.html` certainly looks interesting.



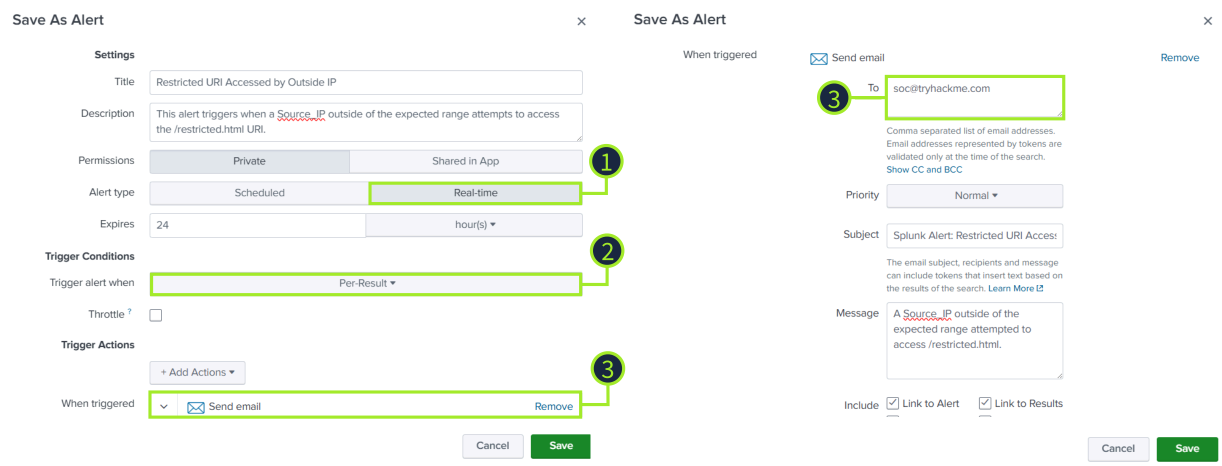
Let's say that `/restricted.html` should normally only be accessed from internal network IP addresses. To identify potential unauthorized access, we can create a query that returns events where the `Source_IP` falls outside the expected internal ranges. If any external IP addresses appear in the results, it could indicate suspicious activity. After entering the query, we can click `Save As` and `Alert`.

```
index = web_logs URI = /restricted.html NOT Source_IP IN (10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16)
```

When saving the alert, you'll need to provide a title and description, and then choose from several configuration options. In the example below, we've selected the following settings:

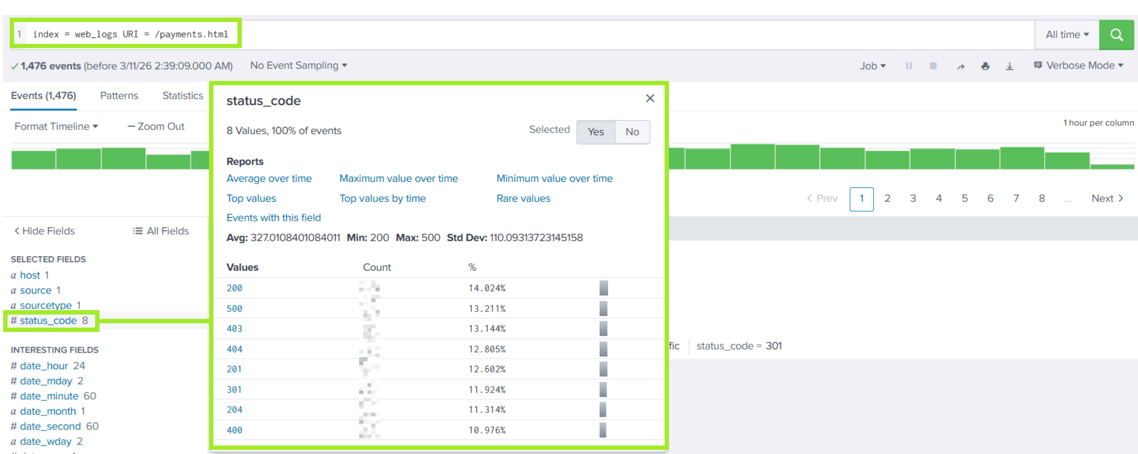
1. **Alert type**: Real-time - Runs continuously and triggers as soon as a matching event appears

2. **Trigger alert when**: Per-Result - Triggers every time a single event matches the search criteria
3. **When triggered**: Send email - An email will be sent to `soc@tryhackme.com` when the alert is triggered



Building a Baseline and Threshold Rule: Now, let's look at another example using the `web_logs` index again from above. We will specifically focus on the `/payments.html` `URI` field. If you query this field in your Splunk instance and examine the `status_code` field, you can see that the page returns a range of responses.

```
index = web_logs URI = /payments.html
```

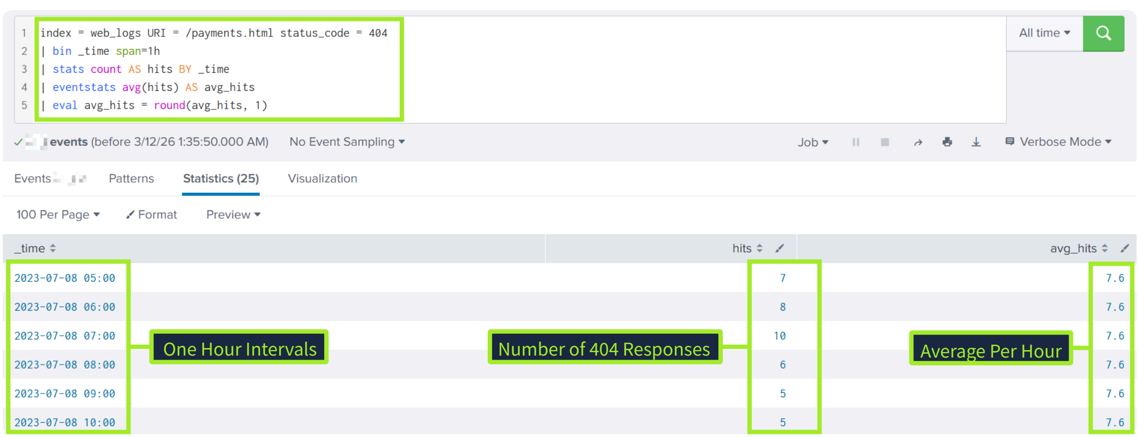


Perhaps we want to build an alert that notifies us if the server has a spike in `404` responses. Well, what do we consider a spike? To answer this, we need to establish a baseline to understand the normal range of this response.

The query below searches the `web_logs` index for `status_code` `404`. It first groups events into 1-hour intervals, counts the number of events, and then calculates the average to one decimal, providing a baseline.

```
index = web_logs URI = /payments.html status_code = 404
| bin _time span=1h
| stats count AS hits BY _time
| eventstats avg(hits) AS avg_hits
| eval avg_hits = round(avg_hits, 1)
```

As shown in the screenshot below, `/payments.html` generates an average of 7.6 `404` responses per hour based on the available events.



Threshold

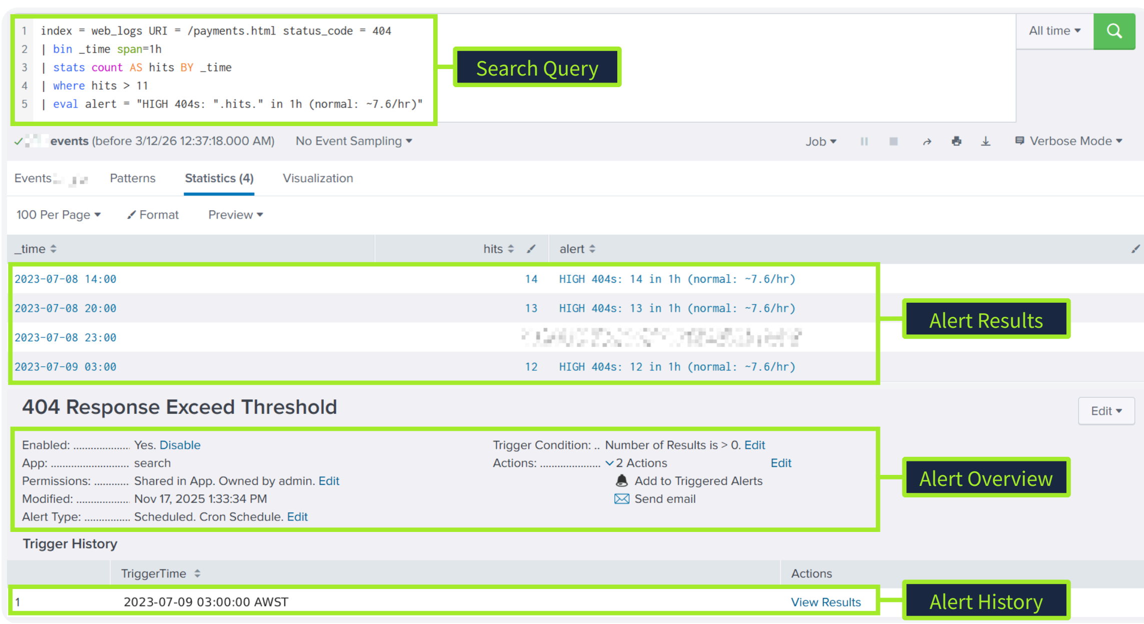
We measured an average of 7.6 `404` responses per hour on `/payments.html`. This is our baseline, but traffic isn't static.

- Weekends, holidays, or marketing campaigns can shift the average
- A fixed number like 10 risks false positives, and a larger number could mean missing real issues

Let's create a query that sets a threshold of `>11` in any 1 hour window. This is approximately 1.45 times the average. From this query, you could create an alert that runs every hour and notifies the SOC team of the abnormal response rate.

```
index = web_logs URI = /payments.html status_code = 404
| bin _time span=1h
| stats count AS hits BY _time
| where hits > 11
| eval alert = "HIGH 404s: ".hits." in 1h (normal: ~7.6/hr)"
```

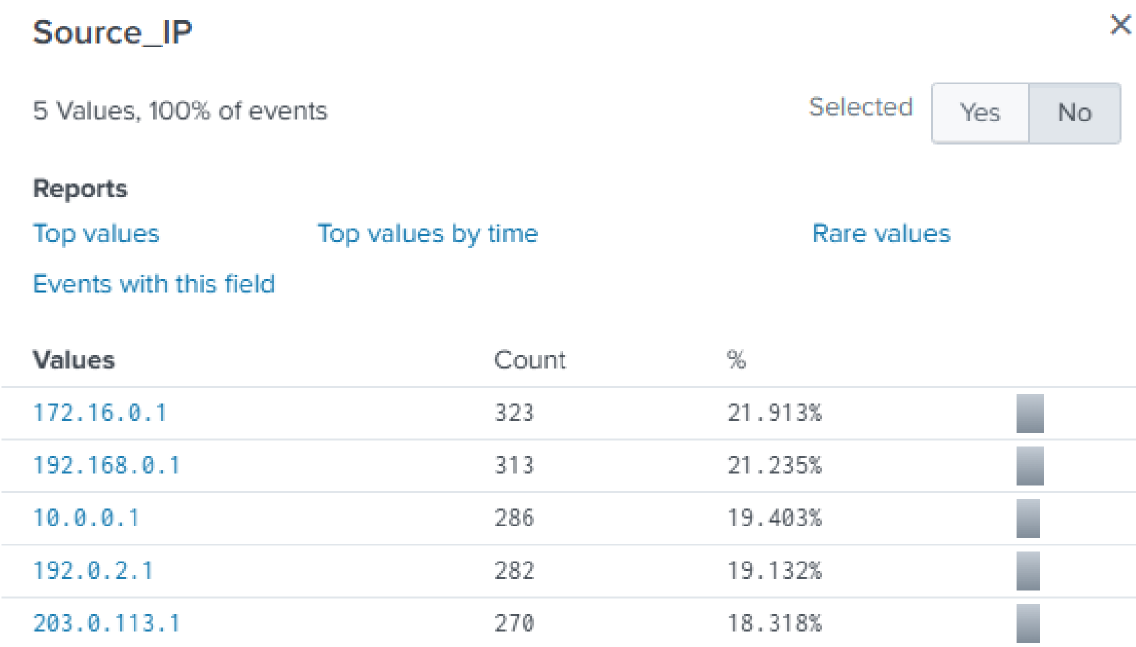
As shown in the screenshot below, if the count exceeds 11, the alert displays the events within the one-hour window, along with the total number of responses received. Just like the previous alert, we can choose to receive a notification in-app or by email, and also configure a variety of additional actions. We can also view the alert's trigger history, allowing us to track when it fired and the results.



Answer the questions below

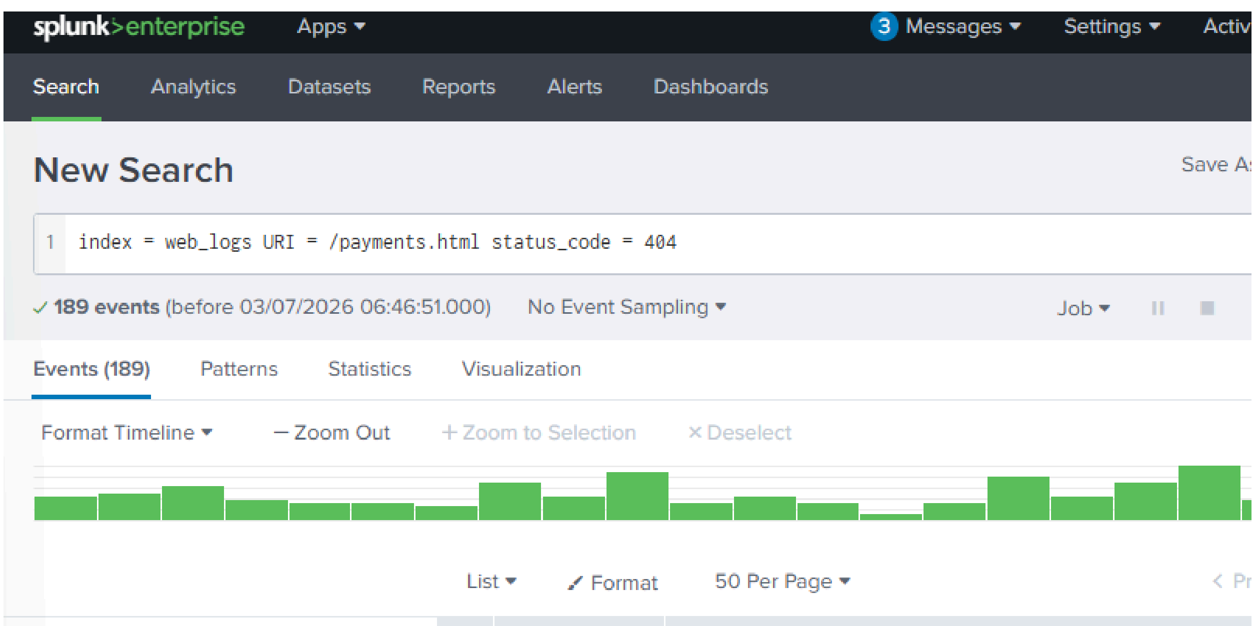
> How many Source_IP addresses outside of the expected range accessed /restricted.html?

2



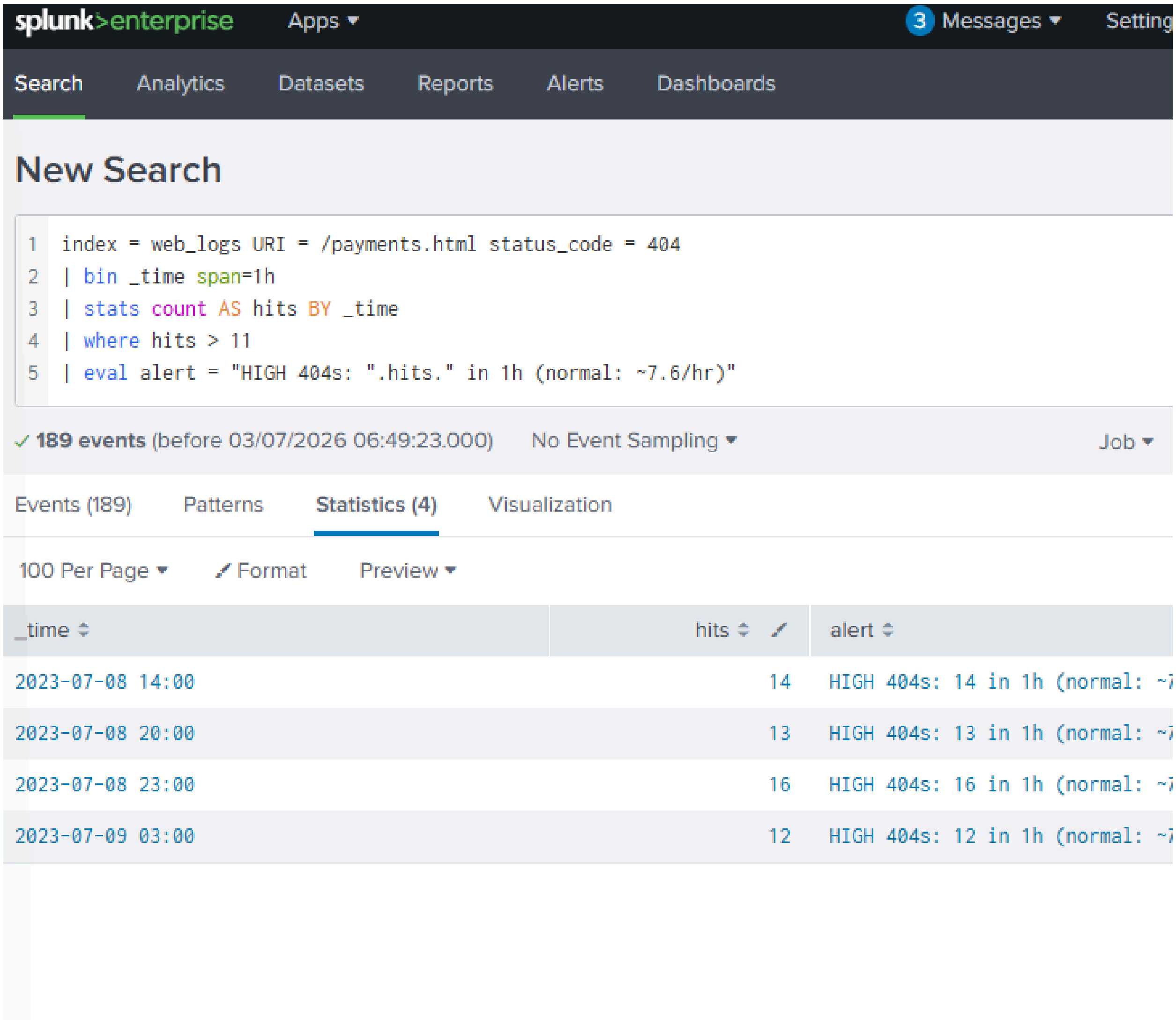
> How many total 404 status_code were recorded at /payments.html?

189



> What is the highest number of 404's received in a given hour with the data available?

16



Among these IP addresses:

IP Address	Internal (Private) or External (Public)
172.16.0.1	Internal (Private)
192.168.0.1	Internal (Private)

IP Address	Internal (Private) or External (Public)
10.0.0.1	Internal (Private)
192.0.2.1	External*
203.0.113.1	External*

Private IP ranges: These ranges are reserved for internal networks (RFC 1918):

- **10.0.0.0 – 10.255.255.255** (`10.0.0.0/8`)
- **172.16.0.0 – 172.31.255.255** (`172.16.0.0/12`)
- **192.168.0.0 – 192.168.255.255** (`192.168.0.0/16`)

Therefore: **172.16.0.1**, **192.168.0.1**, and **10.0.0.1** are **internal/private**.

192.0.2.1 and **203.0.113.1** are not private IPs. They belong to special documentation/test ranges (TEST-NET) used in examples and training material, but in Splunk labs and exam questions they are typically treated as **external/non-internal** addresses.

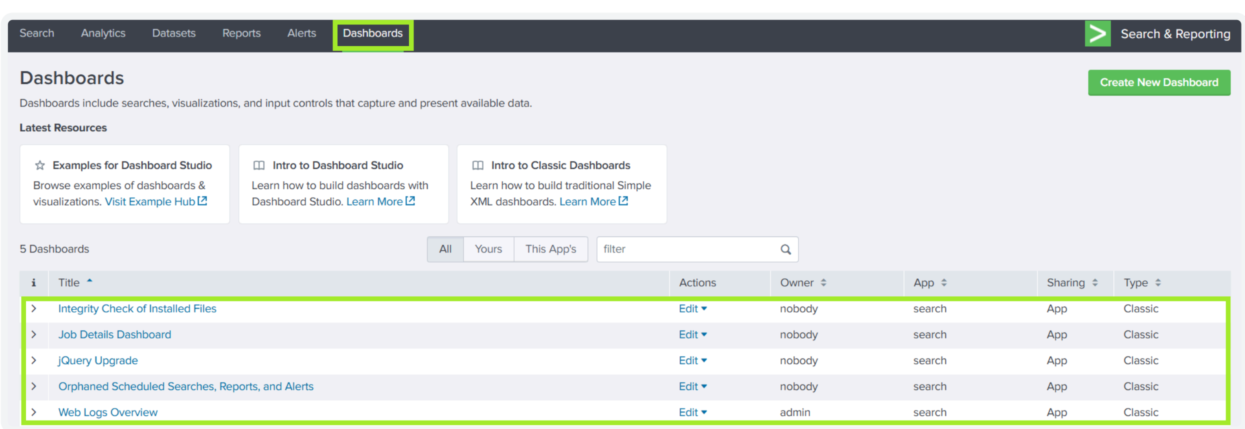
Answer: Internal IPs = **172.16.0.1, 192.168.0.1, 10.0.0.1**.

External/non-internal IPs = **192.0.2.1, 203.0.113.1**.

So there are **2 external IP addresses** in your list.

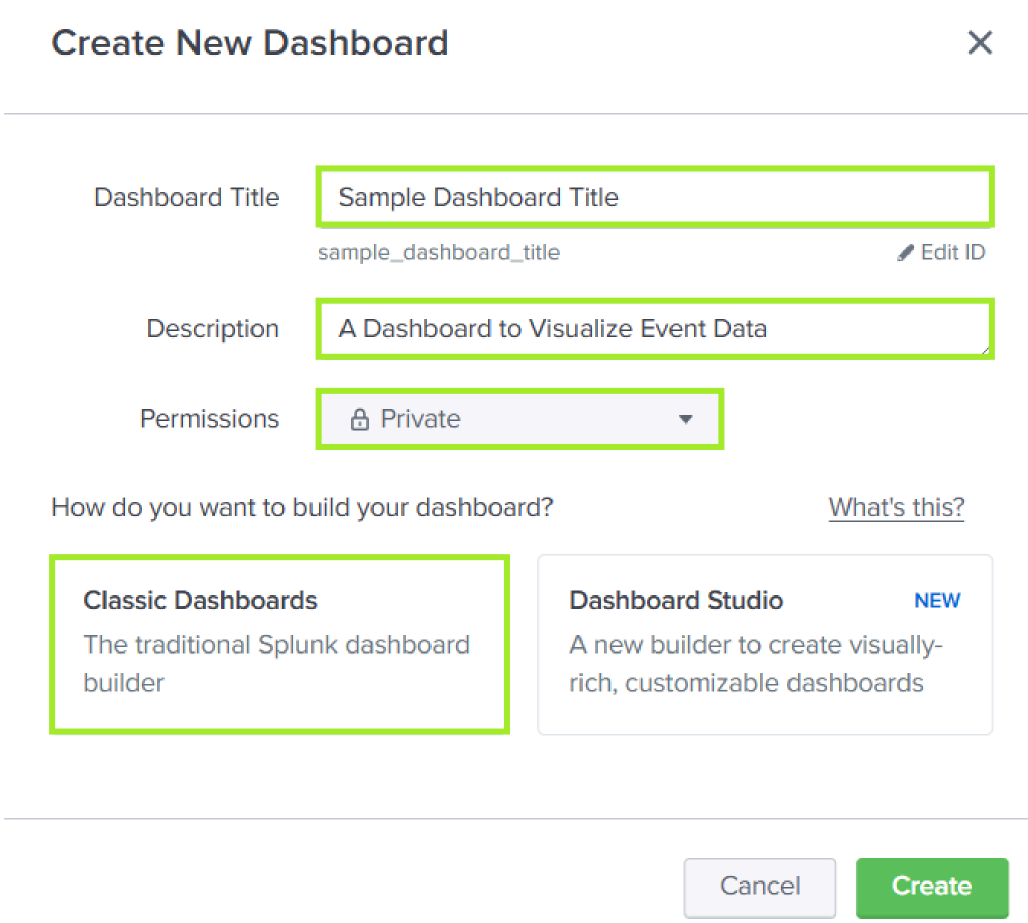
Task 4 - Creating Dashboards for Summarizing Results

If you navigate to the **Dashboards** tab in Splunk, you will see a list of default dashboards provided, as well as the one we will be using in this task: **Web Logs Overview**.



Aside from choosing an existing dashboard, you have the option to create a new one in which you must assign a title, provide an optional description, adjust the permissions, and decide whether to use Classic Dashboards or Dashboard Studio.

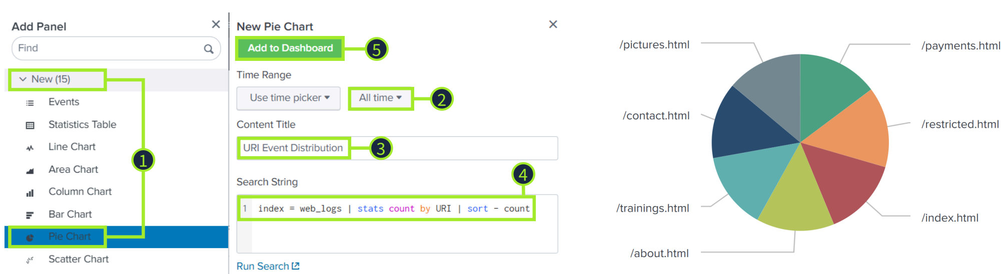
Dashboard Studio is Splunk's newer dashboard builder, designed to provide users with greater customization options in exchange for a complex learning curve. Classic Dashboards, on the other hand, remain the most commonly encountered format and fully support all standard visualizations. In this room, we will cover Classic Dashboards.



Back in the dashboards tab, go ahead and select the **Web Logs Overview** dashboard. Currently, our dashboard has a single panel that visualizes the count of events over time from the `web_logs` index. This dashboard is helpful, but we can further enhance it by adding more visualizations to better understand the data from our web server. Go ahead and click `Edit`, then `+ Add Panel` as highlighted in the screenshot below, so we can expand our current dashboard.

Perhaps, we want to build a pie chart that shows the event count for the `URI` field in the `web_logs` index, helping us visualize how many times each `URI` was accessed within our available events. In the `Add Panel` pop-out, choose:

1. New → Pie Chart
2. Set the time to `All time`
3. Enter a Content Title
4. Enter the search string `index = web_logs | stats count by URI | sort - count`
5. Click **Add to Dashboard**



We can add more panels to display any information we like. In the previous task, we looked at the `/restricted.html` `URI` field. Let's create a stats table for our dashboard that shows:

- `status_code` field

- count of events
- percent of events
- total amount of events overall

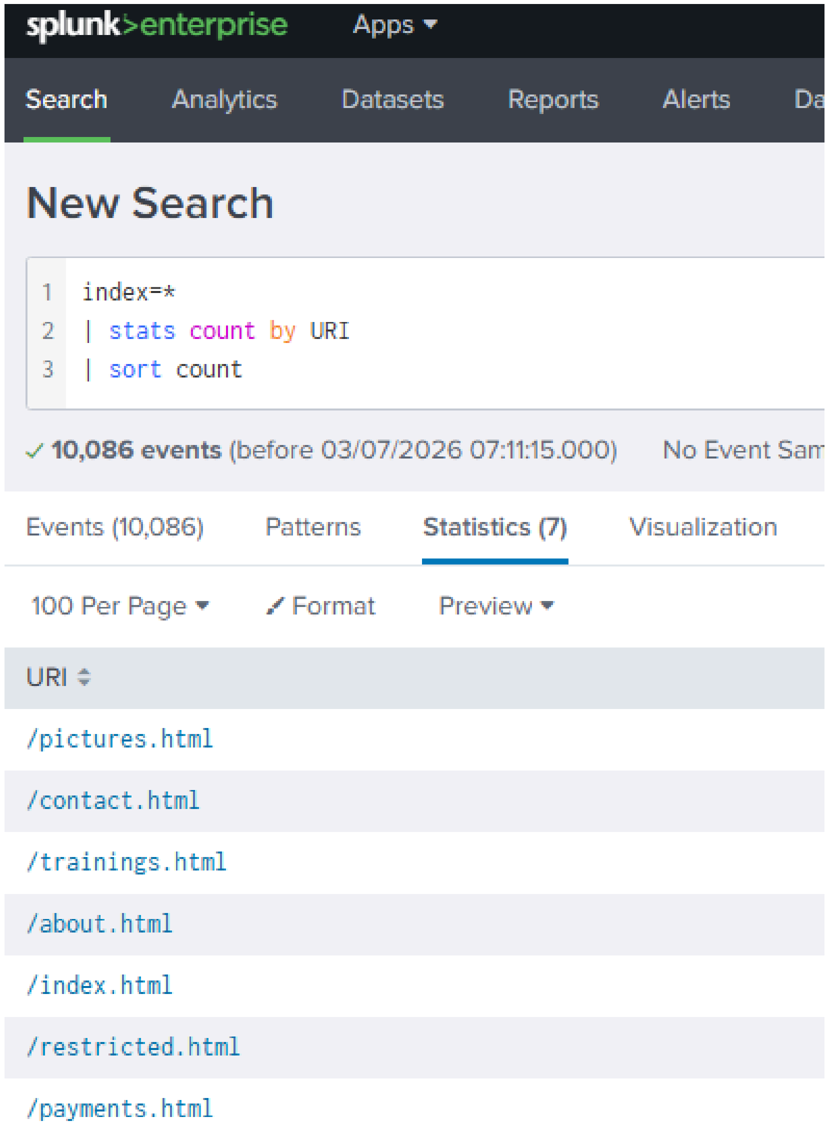
Go ahead and click `+ Add Panel` button as you did previously, but this time, choose New → Statistics Table. Next, set your Time Range to All time and enter the following query as the Search String:

```
index = web_logs URI = /restricted.html
| stats count by status_code
| eventstats sum(count) as total
| eval percent = round(count * 100.0 / total, 2)
| sort - count
```

Answer the questions below

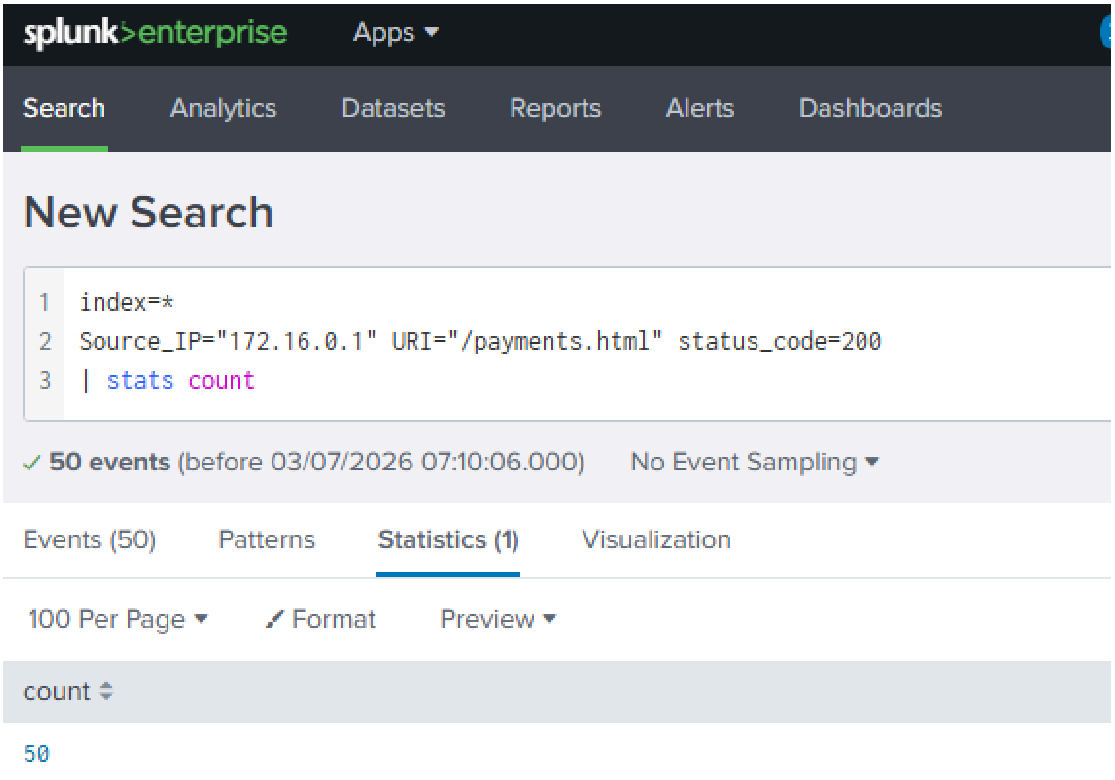
> Inspect the URI pie chart you built in the dashboard above. Which URI field value has the least amount of events present?

/pictures.html



> Add another statistics table to your dashboard to view the Source_IP, URI, and status_code fields. How many times did 172.16.0.1 receive the status_code 200 from /payments.html?

50



This is the hint you're looking for: Try this query: index = web_logs | stats count by Source_IP URI status_code | sort - count

Task 5 - Extending Splunk's Functionality

Splunk Enterprise Security: Provides a complete SOC framework with security visibility, correlation searches, threat intelligence, risk scoring, MITRE ATT&CK mapping, and investigation workflows.

Splunk UEBA: Detects abnormal user and entity behavior by analyzing activity over time and assigning risk scores to identify insider threats, compromised accounts, and suspicious activity.

Splunk SOAR: Automates incident response using playbooks that perform actions such as IP reputation checks, host isolation, and account disabling, reducing response time and repetitive manual work.

Answer the questions below

> Which feature in Splunk Enterprise Security provides visibility into SOC efficiency and performance?

SOC Operations

> What does Splunk SOAR use to automate response actions based on conditions, filters, and decision logic?

Playbooks

Task 6 - Conclusion

Splunk reports and dashboards improve event visibility, while detection rules, ES, and SOAR enable threat detection, correlation, automation, and incident response—turning Splunk into a complete SOC platform.